$$N'_{\mathtt{m}} = \sum_{n \text{ s.t. } a_n \in L_{\mathtt{m}}} \alpha_n; \quad N'_{\mathtt{n}} = \sum_{n \text{ s.t. } a_n \in L_{\mathtt{n}}} \alpha_n \quad (20)$$

where $\alpha_n = P_{\mathtt{U}}(\text{answerable}|q, S^n)$. Therefore, the SelfCheckGPT with QA score, $\mathcal{S}_{\text{QA}}$, is:

$$\mathcal{S}_{\text{QA}} = P(\text{F}|L_{\mathtt{m}}, L_{\mathtt{n}}) = \frac{\gamma_2^{N'_{\mathtt{n}}}}{\gamma_1^{N'_{\mathtt{m}}} + \gamma_2^{N'_{\mathtt{n}}}} \quad (21)$$

In Table 5, we show empically that applying Bayes' theorem and soft counting α (in Equation 20) improves the performance of the SelfCheckGPT with QA method.

Varaint	Sentence-lvl			Passage-lvl	
	NoF	NoF*	Fact	PCC	SCC
SimpleCount	83.97	40.07	47.78	57.39	55.15
+ Bayes	83.04	38.58	47.41	56.43	55.03
+ Bayes + α	84.26	40.06	48.14	61.07	59.29

Table 5: Performance of SelfCheckGPT-QA's variants.

C SelfCheckGPT with Prompt

We use the prompt template provided in the main text (in Section 5.5) for both GPT-3 (text-davinci-003) and ChatGPT (gpt-3.5-turbo). For ChatGPT, a standard system message "`You are a helpful assistant.`" is used in setting up the system.

At the time of conducting experiments, the API costs per 1,000 tokens are \$0.020 for GPT-3 and \$0.002 for ChatGPT. The estimated costs for running the models to answer `Yes/No` on all 1908 sentences and 20 samples are around \$200 for GPT-3 and \$20 for ChatGPT. Given the cost, we conduct the experiments on 4 samples when performing the ablation about LLM choice for SelfCheckGPT-Prompt (Section 7.3). Table 6 shows the breakdown of predictions made by GPT-3 and ChatGPT.

GPT-3 \ ChatGPT	Yes	No
Yes	3179	1038
No	367	3048

Table 6: Breakdown of predictions made by GPT-3/ChatGPT when prompted to answer `Yes`(supported)/`No`(not-supported).

D Additional Experimental Results

Here, we provide experimental results that are complementary to those presented in the main paper.

n-gram	Sent-lvl AUC-PR			Passage-lvl	
	NoFac	NoFac*	Fact	Pear.	Spear.
Avg($-\log p$)					
1-gram	81.52	40.33	41.76	40.68	39.22
2-gram	82.94	44.38	52.81	58.84	58.11
3-gram	83.56	44.64	53.99	62.21	63.00
4-gram	83.80	43.55	54.25	61.98	63.64
5-gram	83.45	42.31	53.98	60.68	62.96
Max($-\log p$)					
1-gram	85.63	41.04	58.47	64.71	64.91
2-gram	85.26	39.29	58.29	62.48	66.04
3-gram	84.97	37.10	57.08	57.34	60.49
4-gram	84.49	36.37	55.96	55.77	57.25
5-gram	84.12	36.19	54.89	54.84	55.97

Table 7: The performance using different n-gram models in the SelfCheckGPT with n-gram method.

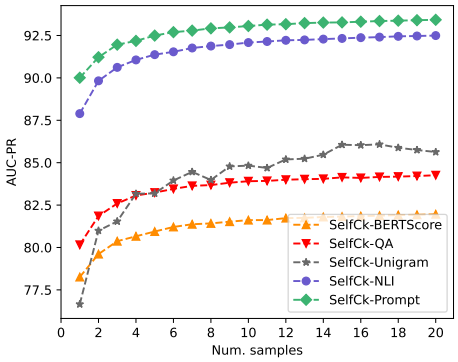


Figure 8: The performance of SelfCheckGPT methods on sentence-level non-factual detection (AUC-PR) versus the number of samples. This Figure extends the passage-level results in Figure 7.

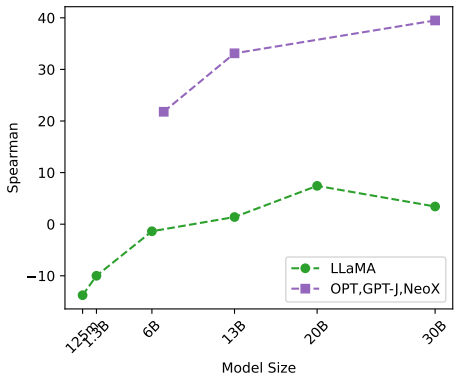
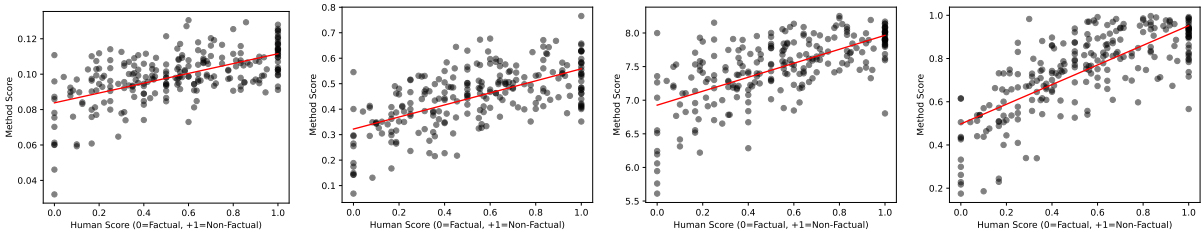


Figure 9: Passage-level ranking performance of the Avg($\mathcal{H}$) method using proxy LLM where the sizes are: LLaMA={7B, 13B, 30B}, OPT={125m, 1.3B, 13B, 30B}, GPT-J=6B, NeoX=20B. The full results are provided in Table 8.



(a) SelfCheckGPT-BERTScore **(b)** SelfCheckGPT-QA **(c)** SelfCheckGPT-1gram(max) **(d)** SelfCheckGPT-NLI

Figure 10: Scatter plot of passage-level scores where Y-axis = Method scores, X-axis = Human scores. Correlations are reported in Table 2. This figure provides results in addition to Figure 6.

LLM	Size	Sentence-level (AUC-PR)			Passage-level (Corr.)	
		NonFact	NonFact*	Factual	Pearson	Spearman
Random	-	72.96	29.72	27.04	-	-
Avg($-\log p$) Method						
LLaMA	30B	75.43	30.32	41.29	21.72	20.20
LLaMA	13B	74.16	30.01	37.36	13.33	12.89
LLaMA	7B	71.69	27.87	31.30	-2.71	-2.59
OPT	30B	67.70	24.43	25.04	-32.07	-31.45
NeoX	20B	69.00	24.38	26.18	-31.79	-34.15
OPT	13B	67.46	24.39	25.20	-33.05	-32.79
GPT-J	6B	67.51	24.28	24.26	-38.80	-40.05
OPT	1.3B	66.19	24.47	23.47	-35.20	-38.95
OPT	125m	66.63	25.31	23.07	-30.38	-37.54
Avg($\mathcal{H}$) Method						
LLaMA	30B	80.80	39.01	42.97	33.80	39.49
LLaMA	13B	80.63	38.98	40.59	29.43	33.12
LLaMA	7B	78.67	37.22	33.81	19.44	21.79
OPT	30B	77.13	33.67	29.55	-0.43	3.43
NeoX	20B	77.40	32.78	30.13	5.41	7.43
OPT	13B	76.93	33.71	29.68	0.25	1.39
GPT-J	6B	76.15	33.29	28.30	-2.50	-1.37
OPT	1.3B	74.05	31.91	26.33	-10.59	-10.00
OPT	125m	71.51	30.88	25.36	-14.16	-13.76
Max($-\log p$) Method						
LLaMA	30B	74.01	27.14	31.08	-22.83	-22.71
LLaMA	13B	71.12	26.78	28.82	-34.93	-31.70
LLaMA	7B	69.57	25.91	26.54	-42.57	-38.24
OPT	30B	67.32	24.40	24.32	-49.51	-45.50
NeoX	20B	67.51	23.88	24.82	-47.96	-44.54
OPT	13B	67.36	24.67	24.46	-50.15	-44.42
GPT-J	6B	67.58	23.94	23.93	-51.23	-47.68
OPT	1.3B	68.16	25.85	24.66	-45.60	-42.39
OPT	125m	69.23	27.66	24.14	-39.22	-37.18
Max($\mathcal{H}$) Method						
LLaMA	30B	80.92	37.32	37.90	35.57	38.94
LLaMA	13B	80.98	37.94	36.01	32.07	34.01
LLaMA	7B	79.65	35.57	31.32	22.10	22.53
OPT	30B	76.58	33.44	29.31	1.63	6.41
NeoX	20B	76.98	31.96	29.13	5.97	9.31
OPT	13B	76.26	32.81	29.25	1.42	2.82
GPT-J	6B	75.30	32.51	28.13	-2.14	1.41
OPT	1.3B	73.79	31.42	26.38	-9.84	-9.80
OPT	125m	71.32	31.65	25.36	-18.05	-17.37

Table 8: AUC-PR for Detecting Non-Factual and Factual Sentences in the GPT-3 generated WikiBio passages. Passage-level PCC and SCC with LLMs used to assess GPT-3 responses. This table is an extension to Table 2.